

Surname

 Initials

Student Number

Stick Barcode Sticker here!

/ 94

FSK116

June Aegrotat Examination Department of Physics University of Pretoria

Date : 26 June 2019
 Examiner : M.J. Legodi
 Moderator : R.Q. Odendaal
 Duration : 3.0 hours
 Final Marks : Up to 94

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 REGISTER: Carefully fold and tear along the dashed line and hand over strip to the invigilator.
 FSK 116, Aegrotat Examination, 26 June 2019, M.J. Legodi.

Surname		Initials	
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Venue			
Signature			

Stick Barcode Sticker here!

June Examinations

Department of Physics, University of Pretoria

Date: 26 June 2019

The following rules apply

All test and examination regulations of the University of Pretoria apply.

Answer all questions in the spaces provided on the question paper. If you need more space, please use the back of the pages and clearly indicate where the question continues.

Pocket calculators may be used, but programmable calculators are not allowed.

No cell phones are allowed in the examination venue.

This paper may not be taken out of the examination venue.

Work written in pencil will not be marked.

Please provide labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct significant figures, etc. Failure to do so may result in marks being deducted.

Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not mean that you will receive full marks.

Units must be used when numbers are substituted else marks will be deducted.

Extra points will be awarded for correctly placed stickers, correct cheat sheets and to high quality or elegant answers.

Constants, conversion factors and some formulae

TO MARKERS:

1. Please indicate in the script where there is a failure / error in the student's logic. Especially if the failure is fatal! Often, the students struggle to understand that getting a correct answer does not necessarily mean that their workings are correct.
2. Students should lose marks in a question where an error is made; but not in subsequent questions (except if the error makes the subsequent questions significantly easier, in which case some marks are lost.)
3. Please note that the total marks are > 94 . A student can only get up to 94 marks. There are a few extra points distributed in the exam making the actual total > 94 marks. **[Same applies to the shorter sup paper at 50]**
4. We have a marking target: Friday 28 June 2019.

$$g = 9.80 \text{ m/s}^2$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin \alpha + \sin \beta = 2 \sin \frac{(\alpha + \beta)}{2} \cos \frac{(\alpha - \beta)}{2}$$

$$c_{ice} = 2220 \text{ J/(kg} \cdot \text{K)}$$

$$L_F = 333 \text{ kJ/kg}$$

Question 1

1.1 (a) Assume the equation $x = At^3 + Bt$ describes the motion of a particular object, with x having the dimension of length and t having the dimension of time. Determine the dimensions of the constants A and B . (3)

(a) $x = At^3 + Bt$
 $[x] = [At^3] + [Bt]$
 The added/summed terms must have the same dimensions of length.
 $\therefore L = [A]T^3 + [B]T$
 $\Rightarrow [A] = L/T^3$ and $[B] = L/T$

(3)

(b) Determine the time derivative of x and the dimensions of the constant A . (2)

(b) $dx/dt = 3At^2 + B$
 $[dx/dt] = [3At^2] + [B] = L/T$
 $= 3 L/T^3 \cdot T^2 + L/T$
 $= L/T$

(2)

1.2 A pyramid has a height of 481 ft, and its base covers an area of 13.0 acres (see accompanying figure). The volume of a pyramid is given by the expression $V = (1/3)Bh$ where B is the area of the base and h is the height. Find the volume of this pyramid in cubic meters. Show all conversion factors in your calculation. (1 acre = 43 560 ft² and 1 ft = 0.305 m). (4)

$V = \frac{1}{3}Bh$
 $= \frac{1}{3}[(13.0 \text{ acres}) \times (43\,560 \text{ ft}^2/\text{acre})] \times 481 \text{ ft}$
 $= 9.08 \times 10^7 \text{ ft}^3$
 $= 9.08 \times 10^7 \text{ ft}^3 \left(\frac{2.83 \times 10^{-2} \text{ m}^3}{1 \text{ ft}^3} \right)$
 $= 2.57 \times 10^6 \text{ m}^3$

(4)

$1 \text{ ft} = 0.305 \text{ m}$
 $\Rightarrow 1 \text{ ft}^3 = 2.83 \times 10^{-2} \text{ m}^3$

Question 2

The coordinates of an object moving in the xy plane vary with time according to the equations $x(t) = -5.00 \sin(\omega t)$ and $y(t) = 4.00 - 5.00 \cos(\omega t)$, where ω is a constant, x and y are in meters, and t is in seconds.

/10

(a) Determine the components of the velocity of the object at $t = 0$ s. (2)

(a)

$$x(t) = -5.00 \sin \omega t \text{ and } y(t) = 4.00 - 5.00 \cos \omega t$$

$$v_x = \frac{dx}{dt} = \frac{d}{dt}(-5.00 \sin \omega t) = -5.00 \omega \cos \omega t$$

and

$$v_y = \frac{dy}{dt} = \frac{d}{dt}(4.00 - 5.00 \cos \omega t) = 5.00 \omega \sin \omega t$$

At $t = 0$

$$\vec{v} = v_x \hat{i} + v_y \hat{j} = (-5.00 \omega \cos 0) \hat{i} + (5.00 \omega \sin 0) \hat{j}$$

$$= -5.00 \omega \hat{i} \text{ m/s}$$

(2)

(b) Determine the components of the acceleration of the object at $t = 0$ s. (3)

(b) We take the time derivative of velocity to find the acceleration.

$$a_x = \frac{dv_x}{dt} = \frac{d}{dt}(-5.00 \omega \cos \omega t) = 5.00 \omega^2 \sin \omega t$$

and

$$a_y = \frac{dv_y}{dt} = \frac{d}{dt}(5.00 \omega \sin \omega t) = 5.00 \omega^2 \cos \omega t$$

At $t = 0$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} = (5.00 \omega^2 \sin 0) \hat{i} + (5.00 \omega^2 \cos 0) \hat{j}$$

$$= 5.00 \omega^2 \hat{j} \text{ m/s}^2$$

(3)

(c) Write expressions for (i) the position vector, (ii) the velocity vector, and (iii) the acceleration vector of the object at any time t . (3)

(c)

$$\vec{r} = x \hat{i} + y \hat{j} = -5.00 \sin \omega t \hat{i} + (4.00 - 5.00 \cos \omega t) \hat{j}$$

(i) or $\vec{r} = [4.00 \hat{j} + 5.00(-\sin \omega t \hat{i} - \cos \omega t \hat{j})] \text{ m}$

(ii) $\vec{v} = 5.00 \omega (-\cos \omega t \hat{i} + \sin \omega t \hat{j}) \text{ m/s}$

(iii) $\vec{a} = 5.00 \omega^2 (\sin \omega t \hat{i} + \cos \omega t \hat{j}) \text{ m/s}^2$

(2)

(d) Describe, in words, what the path followed by the object would be. (2)

(d) Path is a circle of radius 5.00 m centred at $(x, y) \rightarrow (0, 4.00 \text{ m})$

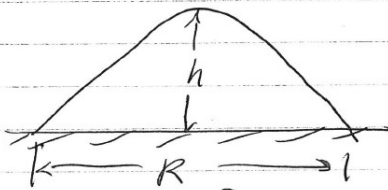
(2)

Question 3

A rock is thrown upwards at an angle θ from level ground in such a way that the maximum height of its flight is equal to its horizontal range R . Ignore any air resistance in this problem.

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(a) At what angle θ is the rock thrown? Do not consider the trivial case where $\theta = 90^\circ$. (3)



$$(a) R = h \rightarrow \frac{v_i^2 \sin 2\theta_i}{g} = \frac{v_i^2 \sin^2 \theta_i}{2g}$$

$$\Rightarrow 2 \sin 2\theta_i = \sin^2 \theta_i \quad \checkmark \quad \text{--- } g \text{ \& } v_i \text{ cancel out}$$

Applying identity;

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\Rightarrow 2(2 \sin \theta_i \cos \theta_i) = \sin^2 \theta_i \quad \checkmark \quad \text{--- } \textcircled{3}$$

$$4 \cos \theta_i = \sin \theta_i$$

$$\tan \theta_i = 4$$

$$\Rightarrow \theta_i = \tan^{-1}(4) = 76.0^\circ$$

(b) In terms of its original range R , what is the range $R_{\max}$ that the rock can attain if it is launched at the same speed but at the optimal angle for maximum range? (2)

$$(b) R_{\text{original}} = \frac{v_i^2 \sin 2\theta_i}{g} = \frac{v_i^2 \sin[2(76.0^\circ)]}{g}$$

$$\text{and } R_{\max} = \frac{v_i^2 \sin[2(45.0^\circ)]}{g}$$

$$\Rightarrow \frac{R_{\max}}{R_{\text{original}}} = \frac{\sin 90^\circ}{\sin 2(76.0^\circ)} = 2.13 \quad \checkmark \quad \textcircled{2}$$

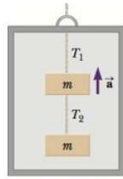
$$\Rightarrow R_{\max} = 2.13 R_{\text{original}} \quad \checkmark$$

(c) Would your answer to part (a) be different if the rock is thrown with the same speed on Mars, for example? Explain. (Take the Marsian gravitational acceleration to be $g_{\text{Mars}} = 3.711 \text{ m/s}^2$.) (2)

(c) The g cancels out in determining (a), therefore the answer will be the same on every planet. $\textcircled{2}$

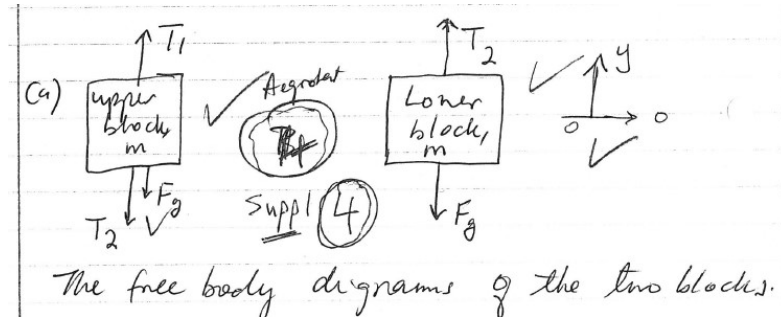
Question 4

4.1 Two blocks, each of mass m , are hung from the ceiling of an elevator as in the figure below. The elevator has an upward acceleration a . The strings have negligible mass.



(a) Sketch the free body diagrams for the masses involved.

(4)



(b) Find the tensions T_1 and T_2 in the upper and lower strings in terms of m , a , and g .

(5)

(b) The acceleration a_y is common for both masses. i.e. $a_y = a$ ✓

Apply Newton II on lower block;

$$\sum F_y = ma_y$$

$$T_2 - F_g = ma$$

$$\Rightarrow T_2 = F_g + ma = m(g + a) \quad (5)$$

Applying Newton II on upper block:

$$\sum F_y = ma_y$$

$$T_1 - T_2 - F_g = ma$$

$$\Rightarrow T_1 = T_2 + F_g + ma$$

$$= mg + ma + mg + ma$$

$$= 2m(g + a) \text{ or } 2T_2$$

(c) Compare the two tensions and determine which string would break first if a is made sufficiently large.

(2)

(c) $T_1 = 2T_2$ ✓ as has been established. As the acceleration of the system is increased, T_1 will break first (2)

(d) What are the tensions if the cable supporting the elevator breaks?

(3)

(d) From $T_1 = 2m(g + a)$

For free fall, $a = -g$

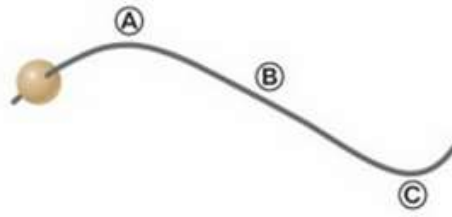
$$\Rightarrow T_1 = 2m(g - g) = 0$$

Since $T_1 = 2T_2$

$$\Rightarrow T_2 = 0 \text{ also} \quad (3)$$

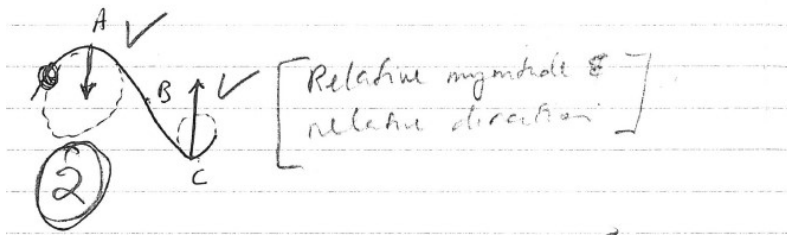
Question 5

5.1 A bead slides at constant speed along a curved wire lying on a horizontal surface as shown in the accompanying figure.



Draw the vectors representing the forces exerted by the wire on the bead at positions A, B and C.

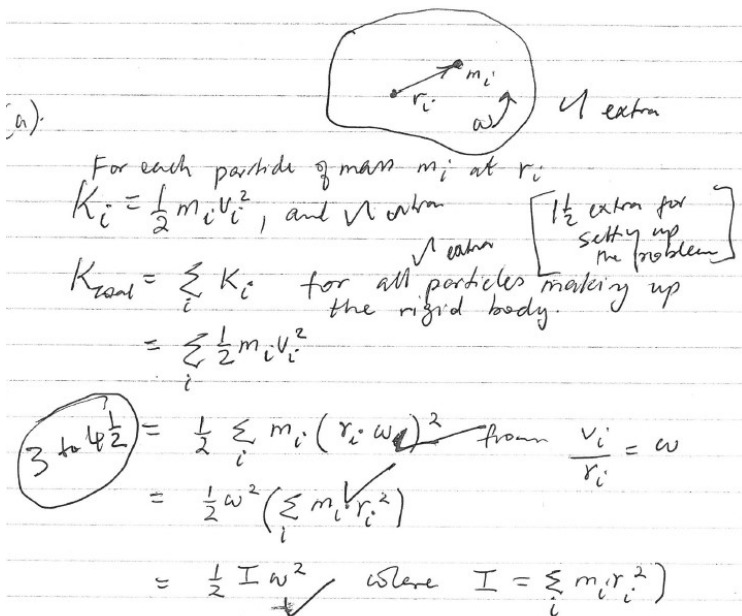
(2)



5.2 Consider a rigid body rotating at a constant angular velocity ω . The body is made up of many particles, each with mass m_i and distance r_i from the rotating axis / axle. The kinetic energy of each of the particles is $K_i = \frac{1}{2} m_i v_i^2$, and the total kinetic energy of the disc is $K = \sum K_i$.

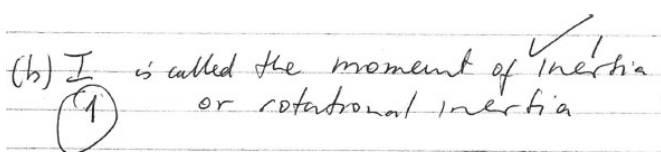
(a) Show that the total kinetic energy of the disc can be written in the form $K = \frac{1}{2} I \omega^2$, and find an expression for I . Show your arguments clearly.

(3)



(b) What is the quantity I called?

(1)



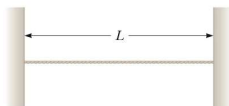
Question 6

6.1 Show that the result of superposition of waves $y_1 = A \sin(kx - \omega t)$ and $y_2 = A \sin(kx - \omega t + \phi)$ are described mathematically as a wave with an amplitude and a phase. Show all your steps! (4)

$$\begin{aligned}
 y' &= y_1 + y_2 \\
 &= A \sin(kx - \omega t) + A \sin(kx - \omega t + \phi) \\
 \sin \alpha \sin \beta &= 2 \sin \frac{(\alpha + \beta)}{2} \cos \frac{(\alpha - \beta)}{2} \\
 \text{let } \alpha &= kx - \omega t \text{ and } \beta = kx - \omega t + \phi \\
 y' &= A \{ \sin[kx - \omega t] + \sin[kx - \omega t + \phi] \} \\
 \therefore y' &= 2A \sin \left(\frac{kx - \omega t + kx - \omega t + \phi}{2} \right) \cos \left(\frac{kx - \omega t - kx + \omega t - \phi}{2} \right) \\
 &= 2A \sin \left[2 \left(\frac{kx - \omega t}{2} \right) + \frac{\phi}{2} \right] \cos \left(-\frac{\phi}{2} \right) \\
 &= 2A \cos \left(-\frac{\phi}{2} \right) \sin \left(kx - \omega t + \frac{\phi}{2} \right) \quad \left(\begin{array}{l} \cos(-\theta) = \cos \theta \\ \text{symbolic} \end{array} \right) \\
 \therefore y' &= 2A \cos \left(\frac{\phi}{2} \right) \sin \left(kx - \omega t + \frac{\phi}{2} \right) \quad \checkmark \\
 &\quad (4)
 \end{aligned}$$

6.2 Two sinusoidal waves of the same amplitude and wavelength are traveling towards one another along a stretched string. Their interference can produce a standing wave pattern described by the equation $y'(x, t) = (2A \sin kx) \cos \omega t$, where the symbols have their usual meanings.

Standing wave patterns can be created by plucking a stretched light string of length L , whose ends are tied between two fixed posts, as shown below.



(a) Derive an expression for the position, x , of the first few nodes and antinodes of the standing wave pattern (i.e. as multiples of λ). (5)

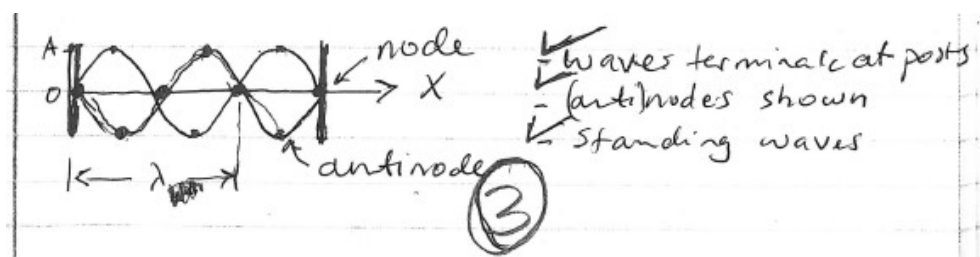
$$\begin{aligned}
 y(x, t) &= (2A \sin kx) \cos \omega t \\
 \text{Nodes:} & \text{ Occur when } \sin kx = 0 \text{ in the amplitude term.} \\
 \text{This happens at } & kx = n\pi, \quad n = 0, 1, 2, 3, \dots \\
 \text{But } kx &= \left(\frac{2\pi}{\lambda} \right) x = n\pi \\
 \text{or } & 2 \times \frac{x}{\lambda} = n \\
 \therefore x &= \frac{n\lambda}{2} \text{ or } n \left(\frac{\lambda}{2} \right) \\
 & \text{for } n = 0, 1, 2, \dots
 \end{aligned}$$

ANTINODES: Occur at amplitude maximum
 i.e. $|\sin kx| = 1$ ✓
 This is true for $kx = (n + \frac{1}{2})\pi$, $n = 0, 1, 2, 3, \dots$ ✓
 Substituting $k = 2\pi/\lambda$
 $\Rightarrow kx = (\frac{2\pi}{\lambda})x = (n + \frac{1}{2})\pi$ ✓
 $= \frac{\lambda}{2\pi} (n + \frac{1}{2})\pi$
 $= \frac{\lambda}{2} (n + \frac{1}{2})$, $n = 0, 1, 2, \dots$
 or
 $x = n(\frac{\lambda}{4})$, $n = 1, 3, 5, \dots$ ✓

3 up to $3\frac{1}{2}$ points
 5 to 6 points

(b) Sketch and label the wave pattern described by your answer in (a) above.

(3)



Question 7

7.1 When θ is small, a simple pendulum motion can be modelled as simple harmonic motion about the equilibrium position, $\theta = 0^\circ$. Consider a mass m attached to the ceiling through a massless and non-stretchable chord. The force of gravity and the tension T are the only forces acting on the pendulum. If the pendulum, at some instant in time, is given a small displacement θ from its equilibrium position, derive an expression for the second order differential equation governing the motion of the pendulum.

(5)

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From: $\sum F = ma$ Newton's II law
The component of the weight, $-mg \sin \theta$ acts as a restoring force, restoring m to its equilibrium position.

$\therefore -mg \sin \theta = ma = m \frac{d^2 s}{dt^2}$ where s is the arc length defined by the motion of the pendulum.
For small angles, $\sin \theta \approx \theta$.
Also, $s = L\theta$ from $\sin \theta = \frac{s}{L}$

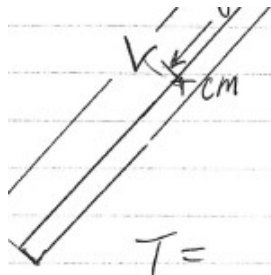
$$\Rightarrow -mg\theta = m \frac{d^2(L\theta)}{dt^2} = mL \frac{d^2\theta}{dt^2}$$

$$\therefore -mg\theta = mL \frac{d^2\theta}{dt^2} \checkmark$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\left(\frac{g}{L}\right)\theta$$

5
Up to
6 points

7.2 A uniform rod of mass M and length L is pivoted about one end and oscillates in a vertical plane. Find the period of oscillation if the amplitude of the motion is small and $\omega = \sqrt{mgd/I}$. (3)



Given $\omega = \sqrt{\frac{mgd}{I}}$

$$T = 2\pi/\omega$$

$$\therefore T = 2\pi \sqrt{\frac{I}{mgd}} \quad \checkmark$$

The rotation is about one end of long rod.

Look up I for long rod or use parallel axis theorem to find I_{rod}

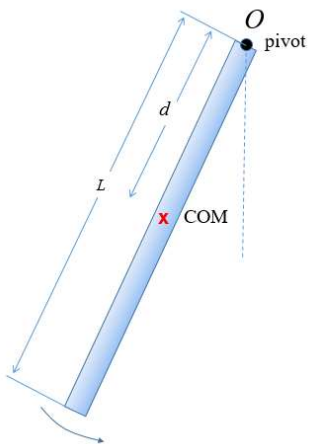
$$I_{rod} = \frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2 = \frac{1}{3}ML^2 \quad \checkmark$$

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$= 2\pi \sqrt{\frac{\frac{1}{3}ML^2}{Mg(\frac{L}{2})}}$$

(3)

$$= 2\pi \sqrt{\frac{2L}{3g}} \quad \checkmark$$



Question 8

8.1 A ball of mass m falls from a height h to the floor.

(a) Reduce Eqn 8.2, the Energy Conservation Equation,

$$\Delta E_{\text{system}} = \sum T \quad (\text{Eqn 8.1})$$

$$\Rightarrow \Delta K + \Delta U + \Delta E_{\text{internal}} = W + Q + T_{\text{MW}} + T_{\text{MT}} + T_{\text{ET}} + T_{\text{ER}} \quad (\text{Eqn 8.2})$$

into the appropriate form for the ball-Earth system and use it to calculate the speed of the ball just before it strikes the Earth. (3.5)

The ball-Earth system is isolated.
 The gravitational energy of the system decreases as the kinetic energy increases.
 (a) $\Delta K + \Delta U = 0$ ✓ Isolated system ✓
 $(\frac{1}{2}mv^2 - 0) + (-mgh - 0) = 0$
 $\Rightarrow \frac{1}{2}mv^2 = mgy$
 $\therefore v = \sqrt{2gh}$ ✓

(b) Re-write Eqn 8.2 into the appropriate version for a ball-only system, and use it to calculate the speed of the ball just before it strikes the Earth. (2.5)

(b) The gravity force does positive work on the ball as it moves downwards.
 The ball is the system and the earth is assumed to remain stationary, so no work is done on it.
 $\therefore W = \Delta K$ ✓
 $\frac{1}{2}mv^2 - 0 = mgh$
 $\Rightarrow \frac{1}{2}mv^2 = mgy$
 $\therefore v = \sqrt{2gh}$ ✓

~~6~~ 6

8.2 A spinning wheel is slowed down by a brake, giving it a constant angular acceleration of 5.60 rad/s^2 . During a 4.20 s time interval, the wheel rotates through 62.4 rad . What is the angular speed of the wheel at the end of the 4.20 s interval? (4)

We will use the following eqns:

$$\theta_f - \theta_i = \omega_i t + \frac{1}{2} \alpha t^2 \quad \text{and}$$

$$\omega_f = \omega_i + \alpha t \quad \text{to obtain:}$$

$$\omega_i = \omega_f - \alpha t \quad \text{and} \quad \theta_f - \theta_i = (\omega_f - \alpha t)t + \frac{1}{2} \alpha t^2$$

(4)

$$= \omega_f t - \frac{1}{2} \alpha t^2$$

Solving for final angular speed:

$$\omega_f = \frac{\theta_f - \theta_i}{t} + \frac{1}{2} \alpha t \quad \checkmark$$

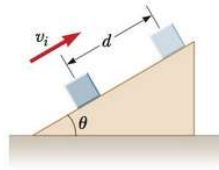
$$= \frac{62.4 \text{ rad}}{4.20 \text{ s}} + \frac{1}{2} (-5.60 \text{ rad/s}^2) (4.20 \text{ s})$$

$$= 3.10 \text{ rad/s}^2 \quad \checkmark$$

Question 9

A 5.00 kg block is set into motion up an inclined plane with an initial speed of $v_i = 8.00 \text{ m/s}$ (see accompanying figure). The block comes to rest after traveling $d = 3.00 \text{ m}$ along the plane, which is inclined at an angle of $\theta = 30.0^\circ$ with respect to the horizontal. For this motion, determine:

/10



(a) The change in the block's kinetic energy.

(2)

Handwritten solution for part (a):

Diagram: A block on an inclined plane at 30.0° . Initial speed $v_i = 8.00 \text{ m/s}$ up the plane. Distance $d = 3.00 \text{ m}$ along the plane. Forces mg and $mg \cos \theta$ are shown.

Consider system: block, plane, Earth, at time instances when the block has just been given its initial speed at at a later instance when it has just stopped.

(a) The change in the kinetic energy of the block

$$\Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$\textcircled{2} = 0 - \frac{1}{2}(5.00 \text{ kg})(8.00 \text{ m/s})^2 \checkmark$$

$$= -160 \text{ J} \checkmark$$

(b) The change in the potential energy of the block-Earth system.

(3)

Handwritten solution for part (b):

(b) The change in the gravitational potential energy is:

$$\textcircled{3} \Delta U = U_f - U_i = (5.00 \text{ kg})(9.80 \text{ m/s}^2)(3.00 \text{ m} \sin 30^\circ)$$

$$= 73.5 \text{ J} (= mgh)$$

(c) The friction force exerted on the block (assumed to be constant).

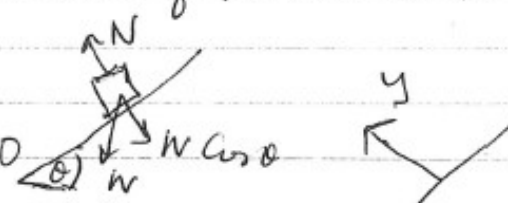
(3)

$$\begin{aligned}
 (c) \quad \sum W_{\text{other forces}} - f_k d &= \Delta K + \Delta U \checkmark \\
 0 - f_k d &= \Delta K + \Delta U \checkmark \\
 \Rightarrow -f_k &= \frac{\Delta K + \Delta U}{d} \\
 \textcircled{3} \quad \therefore f_k &= \frac{(160 \text{ J} - 73.5)}{3.00 \text{ m}} \\
 &= 28.8 \text{ N} \checkmark
 \end{aligned}$$

(d) What is the coefficient of kinetic friction?

(2)

ca) The forces $\perp$ to the incline must add to zero because of Newton's II law.

$$\begin{aligned}
 \sum F_y &= 0 \\
 N - mg \cos \theta &= 0 \\
 \Rightarrow N &= mg \cos \theta
 \end{aligned}$$


But $f_k = \mu_k N$

$$\begin{aligned}
 \therefore \mu_k &= \frac{f_k}{N} = \frac{28.8 \text{ N}}{(5.00 \text{ kg})(9.80 \text{ m/s}^2) \cos 30.0^\circ} \\
 &= 0.679 \checkmark
 \end{aligned}$$

Question 10

- (a) How much heat must be absorbed by ice of mass $m = 720 \text{ g}$ at -10°C to take it to liquid state at 15°C ? (4)

/8

$m = 720 \text{ g}$ at -10°C
to liquid at 15°C

The process could be described in 3 steps:

- ① Raise temperature from -10°C to 0°C } $Q = cm\Delta T$
 $= c_{ice} m (T_f - T_i)$
- ② Phase change at 0°C } $Q = L_f m$
from ice to water
- ③ Raise temperature from 0°C to 15°C } $Q = c_w m \Delta T$
 $= c_w m (T_f - T_i)$

$Q_1 = c_{ice} m (T_f - T_i)$
 $= (2220 \text{ J/kg}\cdot\text{K})(0.720 \text{ kg})[0 - (-10^\circ\text{C})]$
 $= 15.98 \text{ kJ}$ $\Delta T \text{ in } ^\circ\text{C} = \Delta T \text{ in } ^\circ\text{K}$

$Q_2 = L_f m$
 $= (333 \text{ kJ/kg})(0.720 \text{ kg})$
 $= 239.8 \text{ kJ}$

$Q_3 = c_w m (T_f - T_i)$
 $= (4190 \text{ J/kg}\cdot\text{K})(0.720 \text{ kg})(15 - 0)^\circ\text{C}$
 $= 45.25 \text{ kJ}$

$Q_{\text{total}} = Q_1 + Q_2 + Q_3$
 $= (15.98 + 239.8 + 45.25) \text{ kJ}$
 $= 300 \text{ kJ}$

- (b) If we supply the -10°C ice with a total of only 210 kJ as heat, determine the final temperature and phase of the water? (4)

We first need to warm the ice to 0°C

$Q_1 = c_{ice} m \Delta T = c_{ice} m (T_f - T_i)$
 $= (2220 \text{ J/kg}\cdot\text{K})(0.720 \text{ kg})[0 - (-10^\circ\text{C})]$
 $= 15.98 \text{ kJ}$ as previously determined.

Heat still available $= 210 \text{ kJ} - 15.98 \text{ kJ}$
 $= 194 \text{ kJ}$

We then need to determine how much water will be able to melt with the 194 kJ .

$Q_2 = L_f m$
 $\therefore m = Q_2 / L_f$
 $= 194 \text{ kJ} / 333 \text{ kJ/kg}$
 $= 0.580 \text{ kg}$

$\therefore 0.580 \text{ kg}$ is in liquid state at 0°C
and $(0.720 \text{ kg} - 0.580 \text{ kg}) = 0.140 \text{ kg}$
will still be in solid state (ice) at 0°C .